DANCE WAVE - ONLINE DANCE LEARNING PLATFORM

submitted for the partial fulfilment of the requirements for

the award of the degree in

BACHELORS OF COMPUTER APPLICATIONS

BY

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23-UCS-029

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DECLARATION

I, **A. MANOAH (23-UCS-029)** hereby declare that the project report entitled **"ONLINE DANCE LEARNING PLATFORM"** is done under the guidance of **Ms.S.VIJAYA PETERRAJ, B.E., M. C. A., M.Phil.,SET** at UG Computer lab, Loyola College (Autonomous), Chennai-600034 is being submitted in the partial fulfilment of the requirements for the award of the degree in Bachelor of Computer Applications.

DATE:

PLACE: CHENNAI SIGNATURE OF THE STUDENT

BONOFIDE CERTIFICATE

This is to certify that the project work entitled "DANCE WAVE-**ONLINE DANCE LEARNING PLATFORM**" is being submitted to Loyola College (Autonomous), Chennai-600034 by **A.MANOAH (23-UCS-029)** for the partial fulfilment for the award of degree of Bachelor of Computer Applications is a record of work carried out by him, under my guidance and supervision

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DANCE WAVE-ONLINE DANCE LEARNING PLATFORM

Abstract

The **Live Interactive Dance Learning Platform** is a web-based application developed to provide an efficient, flexible, and interactive environment for learning dance through online mediums. In the modern digital era, traditional dance learning methods are limited by geographical constraints, time restrictions, and lack of accessibility to professional instructors. This project aims to overcome these limitations by offering a platform that enables real-time communication between instructors and learners through live video streaming technology.

The system is designed using modern web development technologies such as HTML, CSS, JavaScript, PHP, MySQL, and WebRTC. It provides essential features including user authentication, instructor and student dashboards, class scheduling, and live session management. Instructors can create and manage dance classes, while students can browse, register, and participate in live sessions seamlessly from any location.

One of the key aspects of the platform is its ability to simulate a real classroom environment through interactive live sessions, allowing instructors to provide immediate feedback and guidance to students. This enhances the overall learning experience compared to traditional recorded video-based platforms, which lack interaction and personalization.

The proposed system also ensures scalability, usability, and performance efficiency. The database is structured to handle user information, class schedules, and session data securely, while the user interface is designed to be responsive and user-friendly across various devices including desktops, tablets, and smartphones.

Overall, the Live Interactive Dance Learning Platform serves as an innovative solution that bridges the gap between traditional and digital dance education. It provides an engaging and accessible learning experience, making dance education available to a wider audience while maintaining the quality and effectiveness of real-time instruction. The system not only benefits

students by offering flexible learning opportunities but also empowers instructors to expand their reach and deliver high-quality training globally.

ACKNOWLEDEGEMENT

I extend my deepest gratitude and appreciation to all those who have contributed to the successful completion of my project. This endeavour has been a valuable learning experience, and I am indebted to several individuals and organizations for their support.

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1. INTRODUCTION

1.1 IMPORTANCE OF THE PROJECT

The Live Interactive Dance Learning Platform plays a significant role in transforming the way dance education is delivered and accessed in the modern digital era. Traditional dance learning methods primarily depend on physical presence, where students are required to attend classes at specific locations and fixed schedules. This approach often creates barriers for many learners due to geographical limitations, time constraints, and lack of access to professional instructors. The proposed system addresses these challenges by introducing a web-based platform that enables users to learn dance through live interactive sessions from any location, thereby eliminating the dependency on physical classrooms.

One of the major importance of this project lies in its ability to provide real-time interaction between instructors and students. Unlike pre-recorded video platforms, the system allows learners to actively participate in live classes, ask questions, and receive immediate feedback on their performance. This interactive approach significantly enhances the learning experience, as dance is a practical skill that requires continuous guidance and correction. By enabling real-time communication, the platform ensures that students can improve their techniques effectively and gain a better understanding of dance movements.

Another important aspect of this system is its accessibility and flexibility. The platform allows users to access dance classes at their convenience, without being restricted by location or travel requirements. Students from different regions can connect with professional instructors across the world, creating opportunities for learning diverse dance styles. This not only broadens the learning scope but also promotes cultural exchange through dance. Additionally, the system supports multiple devices such as smartphones, tablets, and computers, making it accessible to a wide range of users.

The project also contributes to reducing time and cost for both students and instructors. Learners no longer need to spend time commuting to dance studios, and instructors can conduct classes without the need for physical infrastructure. This makes the system cost-effective and efficient, benefiting both parties. Furthermore, the platform allows instructors to manage their classes, schedules, and participants digitally, reducing manual work and improving overall productivity

Moreover, the platform supports scalability and future enhancements, allowing the addition of advanced features such as session recording, payment integration, performance analytics, and mobile applications. This ensures that the system can evolve over time to meet changing user needs and technological advancements.

In conclusion, the Live Interactive Dance Learning Platform is an important project that provides a comprehensive solution for online dance education. It enhances accessibility, improves learning quality through real-time interaction, reduces costs, and leverages modern technology to create an efficient and engaging learning environment. The system not only benefits students and instructors but also contributes to the advancement of digital education in the field of dance, making it a valuable and impactful innovation.

1.2. GENERAL ORGANIZATION OF THE REPORT:

This chapter 1 introduces the Live Interactive Dance Learning Platform and explains its importance in the digital era. It highlights how traditional dance learning methods are limited by physical presence and accessibility issues. The chapter describes the need for an online platform that supports real-time interaction between instructors and students. It also outlines the objectives and scope of the project, emphasizing the role of technology in enhancing learning experiences. This section provides a foundation for understanding the system and its relevance in modern education, especially in skill-based learning environments such as dance training.

This chapter 2 focuses on identifying the problems in existing dance learning systems and analysing their limitations. It explains how traditional and recorded video-based platforms fail to provide real-time interaction and personalized feedback. The chapter introduces the proposed system, which offers live interactive sessions to overcome these issues. It also includes system requirements such as hardware and software needed for development. By comparing existing and proposed systems, this chapter highlights the need for an efficient, flexible, and interactive solution that improves accessibility and enhances the overall learning experience.

This chapter 3 describes the design and structure of the system, including architectural design, database design, and graphical user interface design. It explains how different components of the system are organized and interact with each other. The architecture ensures smooth communication between the frontend, backend, and database. The database design focuses on storing user, class, and session data efficiently. The GUI design ensures a user-friendly interface for both instructors and students. This chapter provides a clear blueprint of the system, ensuring that it is scalable, efficient, and easy to use.

This chapter 4 provides a detailed explanation of the system and its functionalities. It describes the various modules such as student module and instructor module. The student module includes registration, login, and joining live classes, while the instructor module includes creating and managing classes. It explains how users interact with the system and how data flows between different components. The chapter also highlights key features such as live video sessions, scheduling, and participation management. This section gives a complete understanding of how the system operates and delivers its intended services.

This chapter 5 explains the development process of the system, including the tools, technologies, and programming languages used. It discusses the use of HTML, CSS, JavaScript, PHP, MySQL, and WebRTC for building the application. The chapter also describes how different components are implemented and integrated. Pseudo code is included to represent the logic behind key functionalities such as login, class creation, and joining sessions. This section provides insight into the technical aspects of the project and demonstrates how the system is built step by step.

This chapter 6 focuses on testing the system to ensure its accuracy, performance, and reliability. It includes various testing methods such as unit testing, integration testing, and acceptance testing. Unit testing verifies individual components, while integration testing checks the interaction between modules. Acceptance testing ensures that the system meets user requirements. Validation is performed to confirm that the system works as intended and produces correct results. This chapter ensures that the system is free from errors and functions effectively under different conditions.

This chapter 7 provides a guide for users to operate the system effectively. It explains how to register, log in, and access different features of the platform. Students can learn how to join live classes, while instructors can understand how to create and manage sessions. The chapter includes step-by-step instructions for using the system, ensuring that users can navigate easily. It also explains system accessibility and basic troubleshooting steps. This section helps users interact with the platform efficiently without technical difficulties.

This chapter 8 explains how the system is implemented in a real-world environment. It includes the process of setting up the server, configuring the database, and deploying the application. The chapter describes how the system is hosted and made accessible to users. It also includes steps for testing, maintenance, and updates after deployment. Deployment ensures that the system runs smoothly and performs efficiently. This section highlights the importance of proper setup and monitoring to maintain system stability and performance.

This chapter 9 summarizes the entire project and highlights its achievements. It explains how the system successfully provides a solution for online dance learning through real-time interaction. The chapter reflects on the benefits of the system, such as improved accessibility, flexibility, and user engagement. It also discusses how the project meets its objectives and addresses the limitations of existing systems. This section provides a final overview of the project and its impact on digital learning.

This chapter 10 discusses potential improvements and additional features that can be added to the system. It suggests enhancements such as class recording, payment integration, mobile application development, and advanced analytics. These improvements aim to increase system functionality and user experience. The chapter highlights the scalability of the system and its ability to adapt to future technological advancements. This section ensures that the system can evolve and remain relevant over time.

This chapter 11 lists all the sources of information used during the project. It includes books, research papers, websites, and technical documentation. Proper references are provided to acknowledge the contributions of various authors and resources. This section ensures academic integrity and helps readers understand the sources of knowledge used in the project.

2. SYSTEM ANALYSIS

2.1 PROBLEM DEFINITION

Traditional dance learning methods are limited by physical presence, fixed schedules, and geographical constraints, making it difficult for many learners to access quality training. Students often face challenges such as lack of nearby instructors, time restrictions, and limited opportunities for personalized guidance. Online platforms that provide recorded videos do not offer real-time interaction or feedback, which is essential for learning dance effectively. This results in reduced engagement and slower skill development. Therefore, there is a need for a system that enables live interaction, flexible learning, and easy accessibility to improve the overall dance learning experience.

2.2 EXISTING SYSTEM

The existing system for dance learning mainly includes offline training centers and online platforms with pre-recorded video tutorials. In offline systems, students must attend physical classes, which can be time-consuming and inconvenient. Online video platforms provide flexibility but lack interaction between instructors and learners. There is no real-time feedback, making it difficult for students to correct mistakes. Additionally, these systems do not provide personalized learning or structured class management. As a result, the existing system is inefficient in delivering an engaging and interactive learning experience, highlighting the need for a more advanced solution.

Disadvantages of Existing System

The existing system of dance learning has several limitations that affect the overall learning experience. One of the major drawbacks is the lack of real-time interaction between instructors and learners, which makes it difficult for students to clarify doubts and receive proper guidance. Since most online platforms provide only pre-recorded videos, there is no immediate feedback or correction, leading to improper learning of dance techniques. Additionally, these systems do not offer personalized learning, as all users receive the same content regardless of their skill level. Offline dance classes also create challenges such as fixed schedules and the need for physical presence, which may not be convenient for all learners. Another significant issue is the absence of a proper progress tracking system, making it difficult for students to monitor their improvement. Furthermore, engagement levels are often low in recorded sessions,

resulting in reduced motivation and interest. Overall, the existing system lacks interactivity, flexibility, and efficiency, making it less effective for skill-based learning like dance.

2.3 PROPOSED SYSTEM

The proposed system is a Live Interactive Dance Learning Platform that enables real-time communication between instructors and students through video streaming technology. It allows instructors to conduct live classes and students to participate from any location. The system includes features such as user registration, class scheduling, live sessions, and feedback mechanisms. It improves learning by providing instant interaction and personalized guidance. The platform is designed to be user-friendly, accessible, and efficient, ensuring better engagement and flexibility. Overall, the proposed system enhances the quality of dance education by overcoming the limitations of traditional and existing methods.

Advantages of Proposed System

The proposed Live Interactive Dance Learning Platform offers several advantages that significantly improve the overall learning experience. One of the major benefits is real-time interaction between instructors and students through live video sessions, which enables immediate feedback and correction of mistakes. This enhances the quality of learning and helps students improve their dance techniques effectively. The system also provides flexibility, allowing users to attend classes from any location at their convenience, thereby eliminating geographical and time constraints. Additionally, the platform includes features such as user registration, dashboards, and class scheduling, which help in better organization and management of learning activities. It also supports a more engaging environment compared to traditional or recorded learning methods, as live sessions increase motivation and participation. Furthermore, the system ensures secure access through authentication mechanisms and maintains structured data for tracking user activities and progress. Overall, the proposed system provides a modern, efficient, and interactive solution for dance education.

2.4 SYSTEM REQUIREMENTS

2.4.1 HARDWARE REQUIREMENTS

- Processor: Intel i3 or above
- RAM: 4GB minimum
- Webcam & Microphone
- Internet connection

2.4.2 SOFTWARE REQUIREMENTS

- HTML, CSS, JavaScript
- PHP
- MySQL
- WebRTC
- Browser (Chrome)

3. SYSTEM DESIGN

3.1 ARCHITECTURE DESIGN:

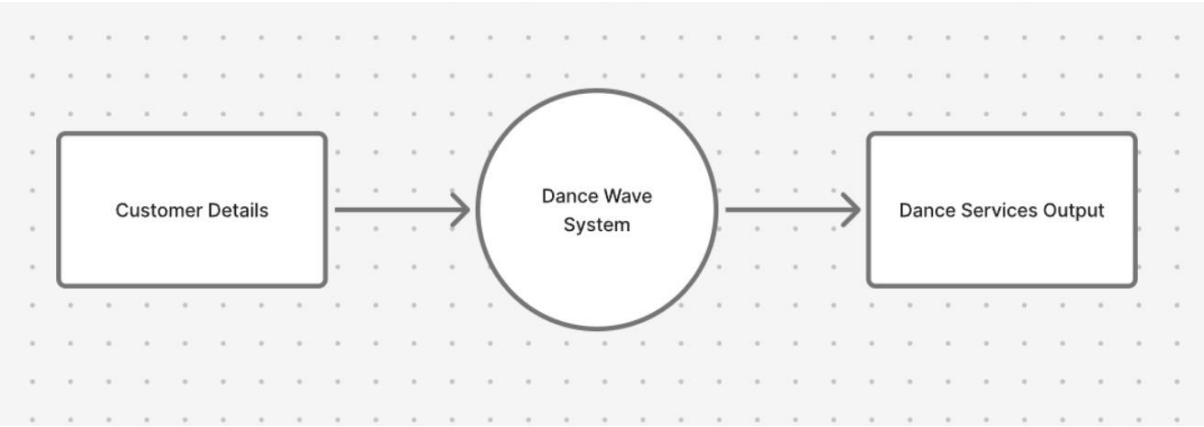
Architectural design describes the high-level structure of the system, including its components and their interactions. The Live Dance Learning Platform follows a client-server architecture where users interact through the frontend, and the backend processes requests and manages data. Technologies such as PHP handle server-side operations, while WebRTC enables real-time video communication. The architecture ensures proper data flow, scalability, and performance. It also separates presentation, business logic, and data layers, making the system more organized, flexible, and easy to maintain.

The system follows client-server architecture:

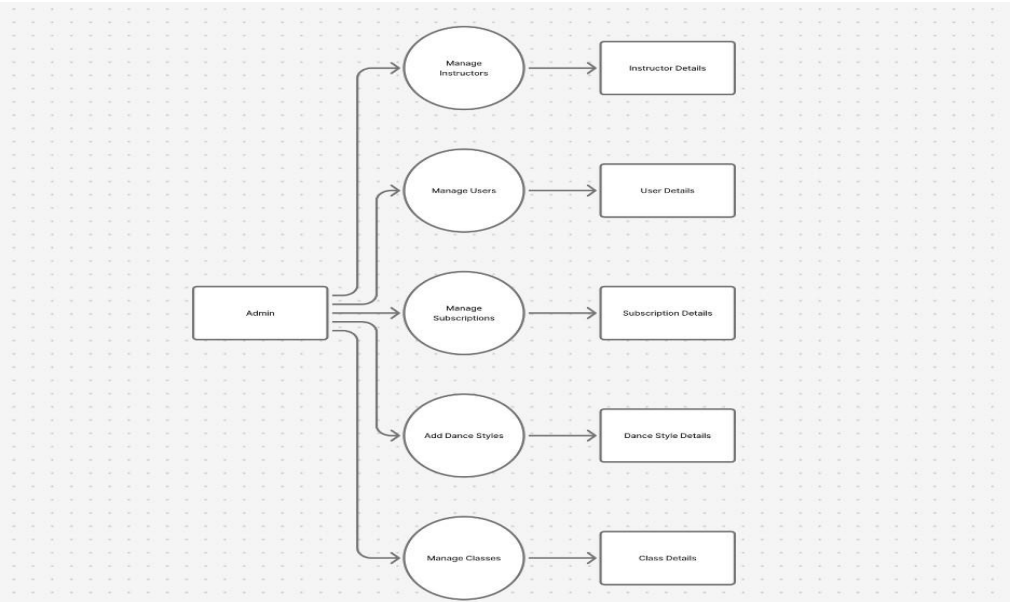
- Frontend → User interaction
- Backend → Processing
- Database → Storage
- WebRTC → Video communication

DFD DIAGRAM:

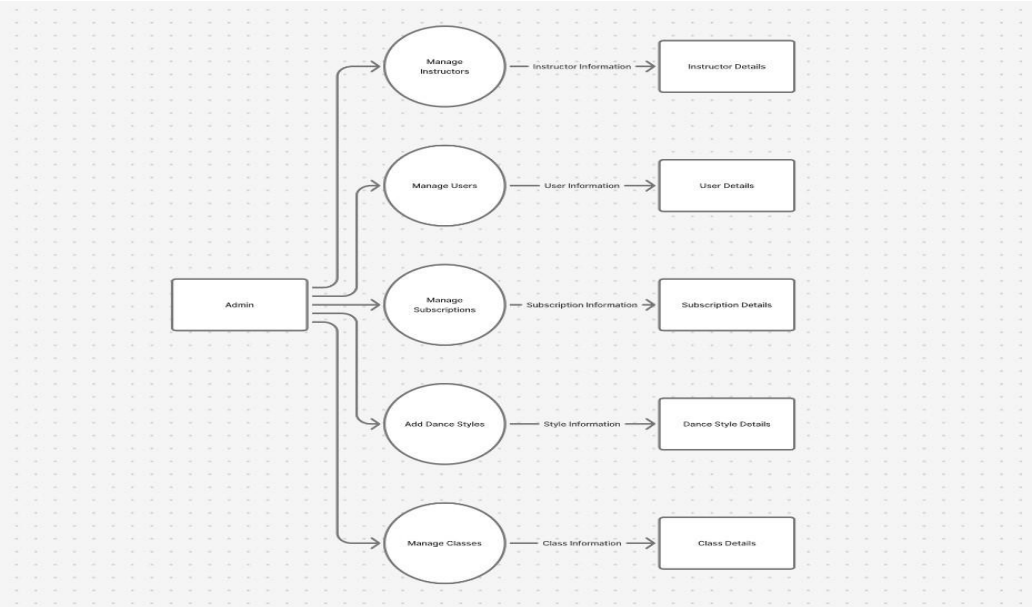
Level 0:



Level1:



Level 2:



3.2 DATABASE DESIGN

Database design focuses on organizing and managing data efficiently to ensure accuracy and consistency. The system uses MySQL to store information such as user details, class schedules, and participation records. Tables like Users, Classes, and Participants are structured with primary and foreign keys to maintain relationships between data. The design supports efficient data retrieval and secure storage. Proper database design reduces redundancy, improves performance, and ensures data integrity, making the system reliable and scalable for handling multiple users and live sessions.

ER DIAGRAM

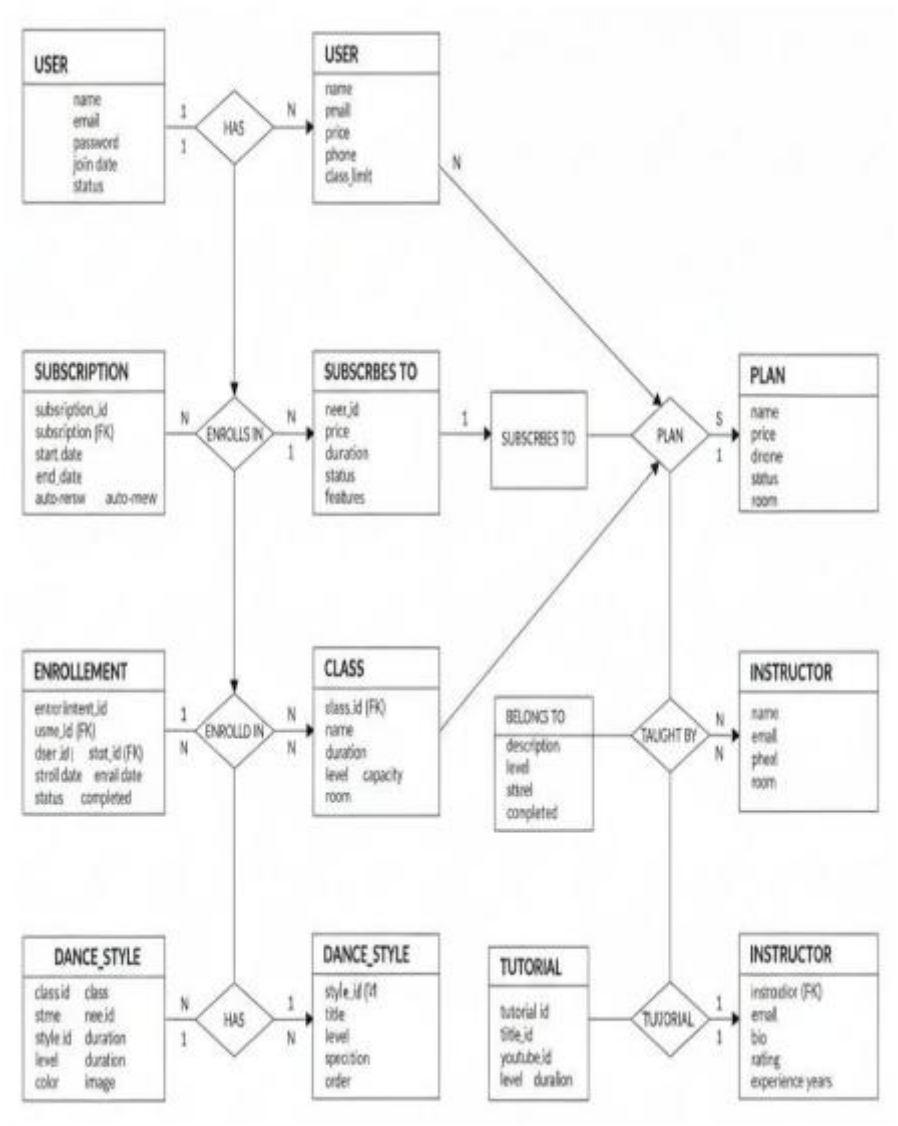


Table Name: Users

Field Name	Type	Null	Constraints
user_id	Int (11)	No	Primary Key
name	Varchar(100)	No	
email	Varchar(100)	No	Unique
password	Varchar(100)	No	
role	Varchar(20)	No	(Student / Instructor)

Table Name: Classes

Field Name	Type	Null	Constraints
class_id	Int (11)	No	Primary Key
instructor_id	Int (11)	No	Foreign Key
class_title	Varchar(100)	No	
description	Text	Yes	
schedule_time	Datetime	No	
meeting_link	Varchar(255)	Yes	

Table Name: Participants

Field Name	Type	Null	Constraints
participant_id	Int (11)	No	Primary Key
user_id	Int (11)	No	Foreign Key
class_id	Int (11)	No	Foreign Key

Table Name: Meetings

Field Name	Type	Null	Constraints
meeting_id	Int (11)	No	Primary Key
class_id	Int (11)	No	Foreign Key
meeting_link	Varchar(255)	No	
created_at	Datetime	No	

Table Name: Feedback

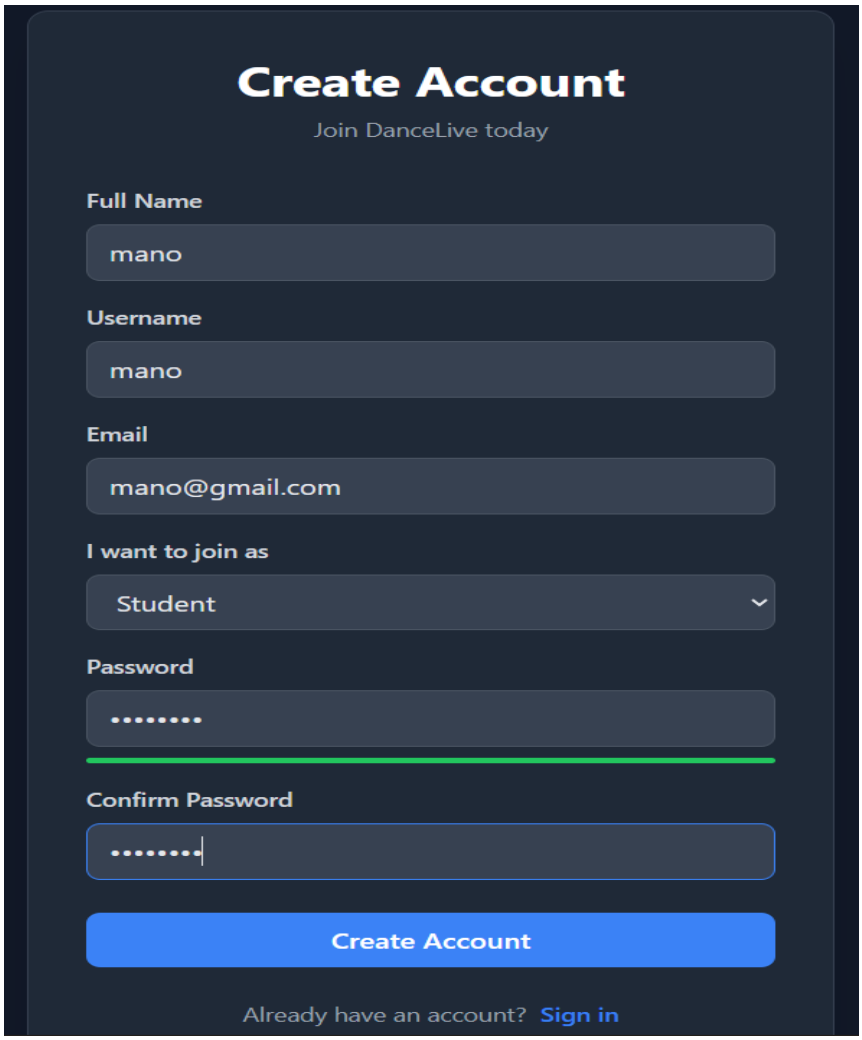
Field Name	Type	Null	Constraints
feedback_id	Int (11)	No	Primary Key
user_id	Int (11)	No	Foreign Key
class_id	Int (11)	No	Foreign Key
comments	Text	No	

Table Name: Payments (Optional Enhancement)

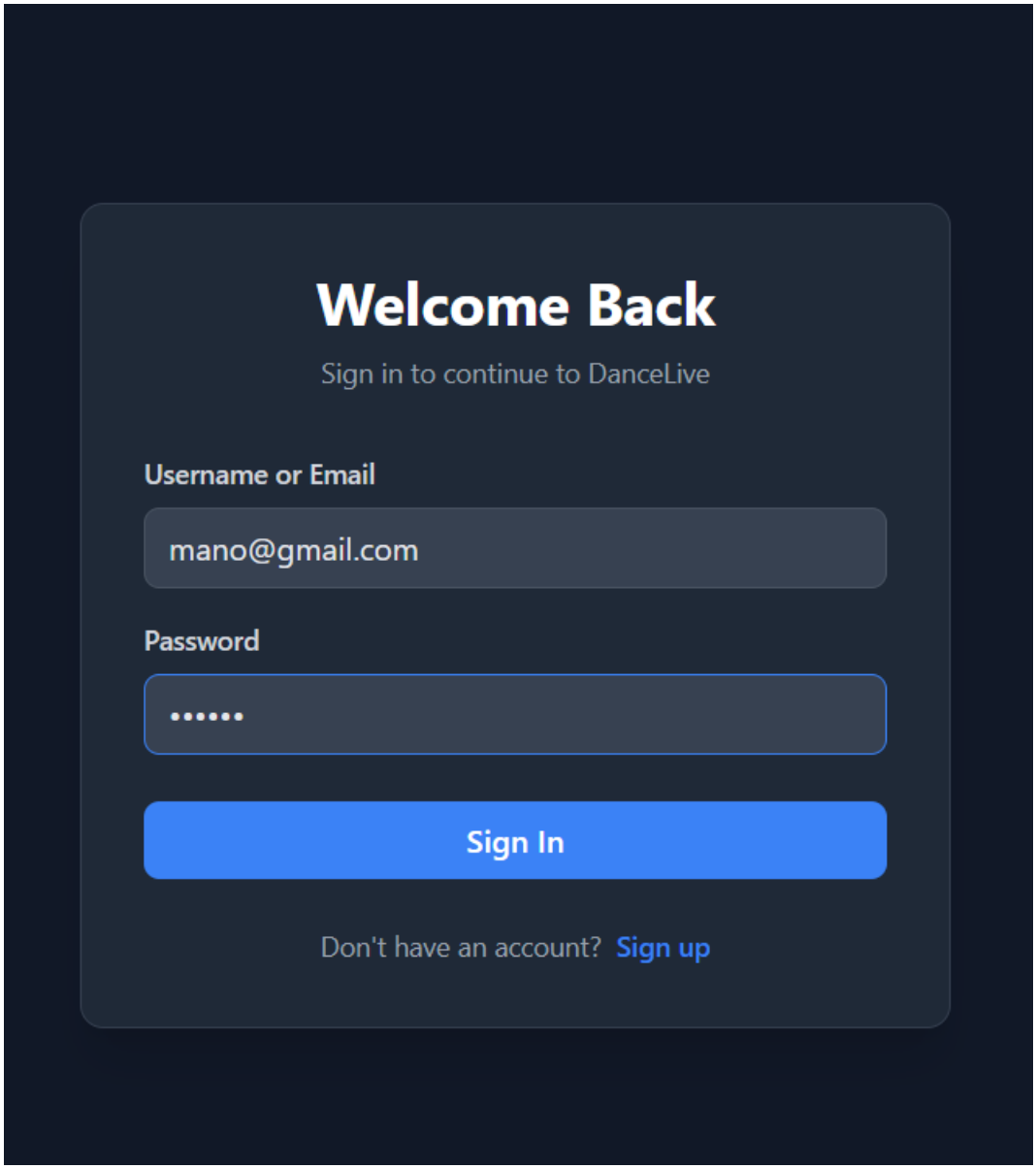
Field Name	Type	Null	Constraints
payment_id	Int (11)	No	Primary Key
user_id	Int (11)	No	Foreign Key
class_id	Int (11)	No	Foreign Key
amount	Decimal(10,2)	No	
payment_status	Varchar(50)	No	
payment_date	Datetime	No	

3.3 GUI DESIGN:

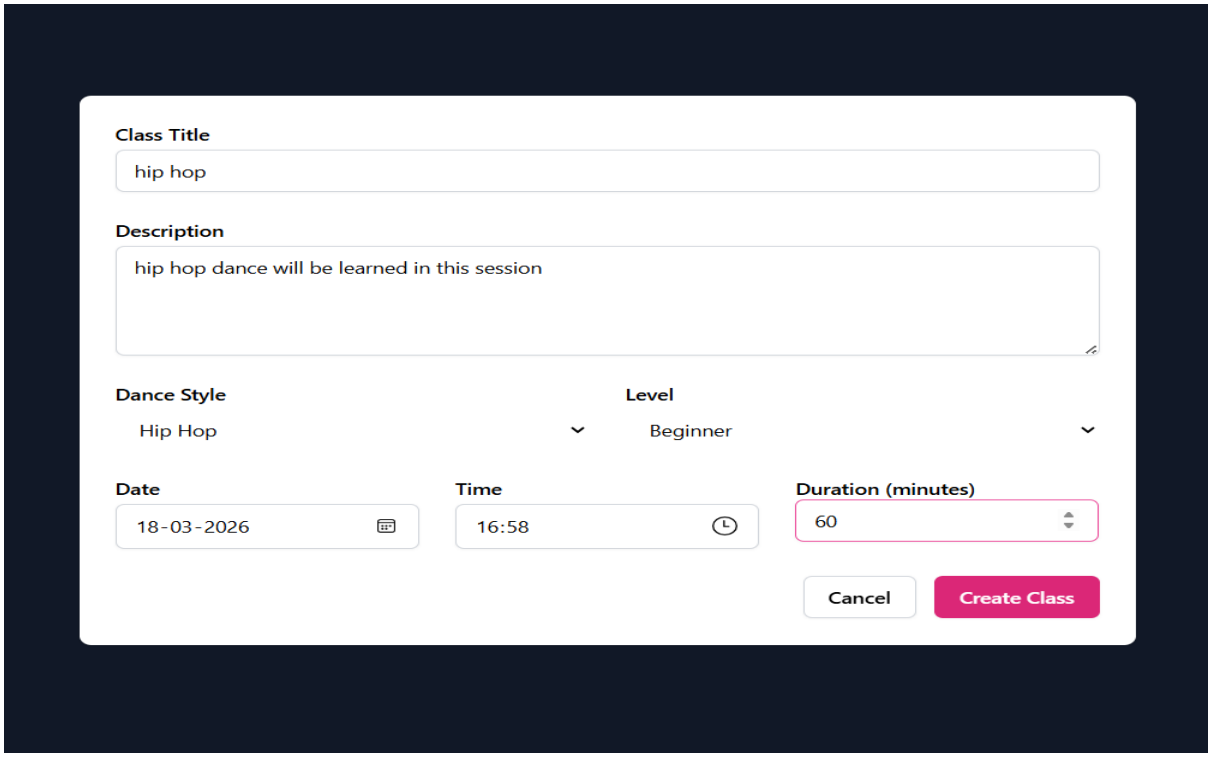
Signup :



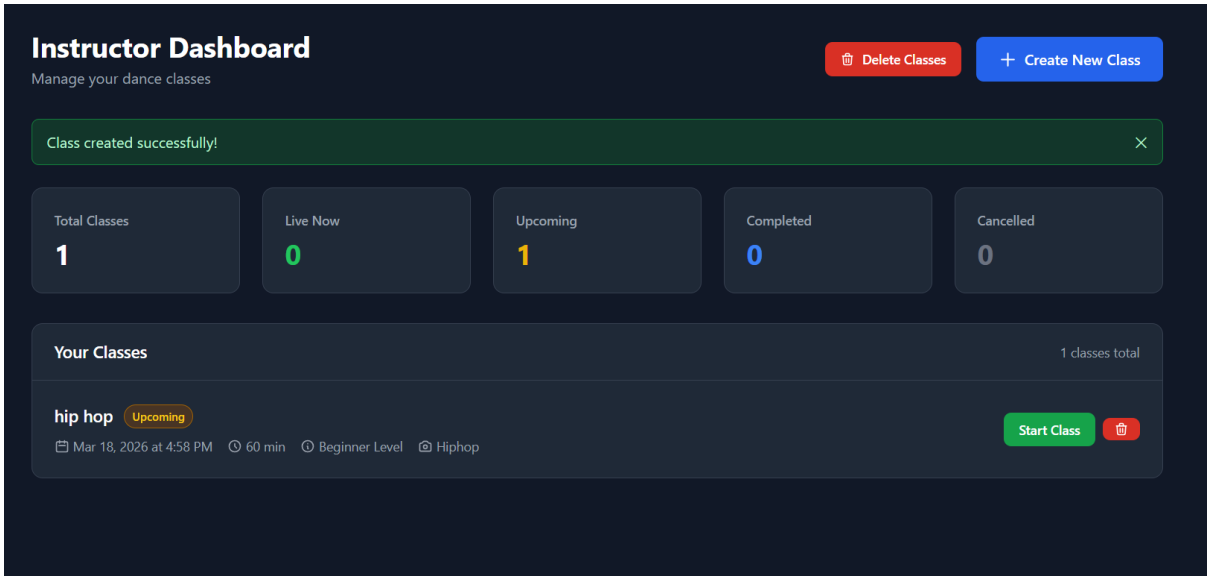
Sign in :


Welcome Back
Sign in to continue to DanceLive
Username or Email
mano@gmail.com
Password
••••••
Sign In
Don't have an account? Sign up

Creating Dance Meeting :



Instructor Dashboard :



4. PROJECT DESCRIPTION

The Live Interactive Dance Learning Platform is a web-based application developed to provide an advanced and interactive environment for learning dance through real-time communication. The system is designed to overcome the limitations of traditional dance learning methods by enabling students and instructors to connect virtually and participate in live dance sessions. By leveraging modern web technologies such as HTML, Tailwind CSS, JavaScript, PHP, MySQL, and WebRTC, the platform creates a seamless and engaging virtual classroom experience. The primary objective of this project is to provide a flexible, accessible, and efficient solution for dance education that allows users to learn from anywhere without geographical constraints.

The system is broadly divided into two major modules, namely the Student Module and the Instructor Module, each designed to perform specific functionalities within the platform. In the Student Module, users are required to register by providing essential details such as name, email, and password. Once registered, they can log in securely using their credentials. After logging into the system, students can browse available dance classes, view schedules, and join live sessions conducted by instructors. The platform ensures that students can easily access live classes through meeting links, providing a smooth and uninterrupted learning experience. Additionally, students can track their participation and improve their skills through continuous practice and interaction.

The Instructor Module is designed to enable instructors to efficiently manage and conduct dance classes. Instructors can log in to the system and create new classes by specifying details such as class name, schedule time, and session links. They can organize both individual and group sessions depending on the requirements. The platform allows instructors to conduct live dance classes using real-time video communication powered by WebRTC technology. During the session, instructors can demonstrate dance movements, observe students' performance, and provide immediate feedback, thereby improving the overall learning experience. This real-time interaction closely replicates the environment of a physical dance classroom.

The system also incorporates a secure authentication mechanism to ensure that only authorized users can access the platform. Role-based access control is implemented to differentiate between student and instructor functionalities. The backend of the system, developed using PHP, handles all business logic including user authentication, class management, and session handling. The MySQL database is used to store and manage data such as user details, class

schedules, and participation records. Efficient database design ensures quick data retrieval and secure storage of information.

One of the key features of the system is the integration of real-time video communication using WebRTC. This technology enables peer-to-peer communication between users, allowing live streaming of video and audio with minimal delay. A signaling mechanism is used to establish connections between participants, ensuring smooth communication during live sessions. The system also supports multiple users joining a session, making it suitable for both small and large classes. This feature significantly enhances user engagement and provides an immersive learning experience.

The graphical user interface (GUI) of the platform is designed using Tailwind CSS to ensure a modern, responsive, and user-friendly layout. The interface includes various pages such as registration, login, dashboard, class management, and live session screens. The responsive design ensures that the platform can be accessed across different devices including desktops, tablets, and mobile phones. The intuitive design makes it easy for users to navigate through the system and access its features without difficulty.

In addition to core functionalities, the system focuses on performance, scalability, and security. Input validation and secure coding practices are implemented to prevent unauthorized access and data breaches. The system is capable of handling multiple users simultaneously, ensuring stable performance during live sessions. The modular design of the application makes it easy to maintain and extend in the future.

Overall, the Live Interactive Dance Learning Platform provides a comprehensive solution for online dance education by combining real-time communication, efficient data management, and a user-friendly interface. It eliminates the limitations of traditional learning methods and offers a flexible and interactive approach to skill development. The system not only enhances accessibility but also improves the quality of learning by enabling direct interaction between instructors and students. With its scalable architecture and modern technology stack, the platform has the potential to be further developed into a full-fledged commercial application for online dance training.

MODULE DESCRIPTION:

The Live Interactive Dance Learning Platform is designed with two primary modules, namely the Student Module and the Instructor Module, to ensure efficient interaction between learners and trainers. The Student Module is developed to provide users with a simple, user-friendly, and interactive environment to access dance learning resources. In this module, students can begin by registering themselves in the system by entering essential details such as their name, email address, and password. Once registered, they can securely log in to the platform using their credentials. After successful authentication, students gain access to the dashboard where they can view a list of available dance classes offered by various instructors. The system enables students to choose classes based on their interests, preferences, and availability. Furthermore, students can join scheduled sessions and actively participate in live classes conducted by instructors. The live session feature allows real-time interaction, enabling students to observe demonstrations, follow instructions, and receive immediate feedback, thereby enhancing their learning experience. This module ensures flexibility and convenience, as students can attend classes from any location with an internet connection, making the learning process more accessible and engaging.

On the other hand, the Instructor Module is designed to facilitate instructors in managing and delivering their dance classes effectively. Instructors can create new classes by providing relevant details such as the class name, type of dance, and a brief description to inform students about the content of the session. Additionally, instructors have the ability to schedule classes by specifying the date and time, allowing students to plan and enroll in advance. At the scheduled time, instructors can conduct live sessions through real-time video communication, which enables direct interaction with students. During these sessions, instructors can demonstrate dance movements, guide students step-by-step, and provide personalized feedback to improve their performance. This module also helps instructors organize their teaching activities systematically and manage multiple sessions efficiently. Overall, both modules work together to create a comprehensive and interactive learning platform that enhances the quality of dance education by combining accessibility, real-time communication, and effective session management.

5. SYSTEM DEVELOPMENT

5.1 LANGUAGE / TOOL

The development of the Live Interactive Dance Learning Platform involves the use of various programming languages and tools that together enable the creation of a scalable, efficient, and user-friendly application. These technologies play a crucial role in designing, developing, and maintaining the system. The combination of front-end, back-end, and database technologies ensures seamless communication between different components of the system. The selected tools provide flexibility, performance, and ease of development, allowing the application to meet both functional and non-functional requirements effectively. By utilizing modern web technologies, the system achieves a responsive design, real-time communication capability, and secure data management, making it suitable for real-world deployment.

5.1.1 HTML (Hypertext Markup Language)

HTML serves as the fundamental building block of the Live Interactive Dance Learning Platform by defining the structure and layout of web pages. It is used to create essential interface components such as forms, buttons, input fields, and navigation elements that allow users to interact with the system. HTML provides a semantic structure that improves readability and accessibility while ensuring compatibility across different browsers and devices. It enables developers to organize content in a logical manner, which enhances the overall usability of the application. Due to its simplicity and ease of use, HTML plays a vital role in rapidly developing the front-end interface of the system.

5.1.2 CSS / Tailwind CSS

CSS and Tailwind CSS are used to enhance the visual presentation and layout of the platform. These technologies allow developers to style web pages by applying colors, fonts, spacing, and positioning to different elements. Tailwind CSS, in particular, provides a utility-first approach that helps in building modern and responsive user interfaces efficiently. The use of responsive design techniques ensures that the application adapts seamlessly to different screen sizes, including mobile devices, tablets, and desktops. CSS also supports animations and transitions, which improve user engagement by creating visually appealing interfaces. Overall, CSS plays a key role in delivering a clean, attractive, and user-friendly experience.

5.1.3 JavaScript

JavaScript is used to add interactivity and dynamic behavior to the platform. It enables real-time updates, event handling, and manipulation of web page elements through the Document Object Model (DOM). In the Live Interactive Dance Learning Platform, JavaScript is particularly important for implementing client-side validation, ensuring that user inputs are correct before being sent to the server. It also facilitates asynchronous communication with the backend using technologies such as AJAX, allowing data to be updated without reloading the entire page. Furthermore, JavaScript is used in integrating WebRTC for real-time video communication, enabling live interaction between instructors and students. This significantly enhances the overall functionality and user experience of the system.

5.1.4 PHP (Hypertext Preprocessor)

PHP is used as the server-side scripting language to handle the backend operations of the system. It is responsible for processing user requests, managing sessions, and implementing business logic. PHP interacts with the database to store and retrieve information such as user details, class schedules, and participation records. It ensures secure authentication by validating user credentials and controlling access based on roles such as student and instructor. PHP also supports integration with web servers, making it suitable for deployment in various environments. Its simplicity, flexibility, and strong community support make it an ideal choice for developing dynamic web applications like this platform.

5.1.5 MySQL

MySQL is used as the database management system to store and organize all the data required for the platform. It maintains structured information such as user accounts, class details, and meeting records. MySQL ensures data integrity and consistency through the use of relational tables, primary keys, and foreign keys. It supports efficient data retrieval through indexing and optimized queries, which improves the performance of the application. Additionally, MySQL provides robust security features, including access control and data encryption, ensuring that sensitive information is protected. Its scalability allows the system to handle an increasing number of users and data without performance degradation.

5.1.6 WebRTC (Real-Time Communication Technology)

WebRTC is a key technology used in the Live Interactive Dance Learning Platform to enable real-time video and audio communication between users. It allows peer-to-peer connection

directly between browsers without requiring additional plugins. WebRTC is used to create live virtual classrooms where instructors and students can interact seamlessly. It utilizes signaling mechanisms to establish connections and supports technologies such as STUN and TURN servers for network traversal. This ensures stable communication even in complex network environments. By integrating WebRTC, the system provides an immersive and interactive learning experience that closely resembles physical dance classes.

5.2 PSEUDO CODE

5.2.1 User Authentication Module

```
Start

User enters username and password

System receives input

Verify user credentials from database

If credentials are valid

    Create user session

    Redirect to dashboard

Else

    Display error message

End
```

5.2.2 Dashboard Loading Module

```
Start

User successfully logs in

Load user dashboard

Fetch available classes from database

Display class list
```

Initialize user interface

End

5.2.3 Class Creation Module (Instructor)

Start

Instructor logs into system

Instructor enters class details (name, time)

System validates input

If input is valid

    Generate meeting link

    Store class details in database

    Display success message

Else

    Display error message

End

5.2.4 Join Class Module

Start

User selects a class

System retrieves class details

If class exists

    Fetch meeting link

    Redirect user to live session

Else

    Display "Class not available"

End

5.2.5 Live Video Session Module

Start

User joins session

Access camera and microphone

Initialize WebRTC connection

Exchange connection details

If connection established

    Start video and audio streaming

    Display participants video

Else

    Display connection error

End

5.2.6 Signup Module

Start

User enters name, email, password

System validates input

If input is valid

    Store user details in database

    Display "Registration successful"

    Redirect to login page

Else

    Display error message

End

5.2.7 Logout Module

Start

User clicks logout

Destroy session

Clear session data

Redirect to login page

End

5.2.8 Database Management Module

Start

System connects to database

If connection successful

    Perform operations (insert, update, delete, fetch)

Else

    Display database error

End

6. SYSTEM TESTING AND VALIDATION

6.1 UNIT TESTING

Unit testing is the process of testing individual components or modules of the system to ensure that each part functions correctly. In the Live Interactive Dance Learning Platform, unit testing is performed on modules such as login, signup, class creation, and database operations. Each module is tested independently with valid and invalid inputs to verify that it produces the expected output. This helps in identifying errors at an early stage and ensures that each component works as intended before integration.

6.2 INTEGRATION TESTING

Integration testing involves combining different modules of the system and testing them as a group to ensure that they work together properly. In this system, modules such as user authentication, class management, and live session handling are integrated and tested. The focus is on verifying data flow between modules, such as ensuring that after login, the user is correctly redirected to the dashboard and can join classes. This testing ensures smooth communication between components and identifies any interface-related issues.

6.3 ACCEPTANCE TESTING

Acceptance testing is carried out to verify whether the system meets user requirements and expectations. In the Live Interactive Dance Learning Platform, this testing ensures that both students and instructors can use the system easily and effectively. Users test functionalities such as registering, logging in, creating classes, and joining live sessions. The system is evaluated based on usability, performance, and reliability. If the system satisfies all requirements, it is considered ready for deployment.

6.4 TEST CASE

1. User Registration Test Case

Test Case ID	TC_01
Test Case Name	User Registration
Description	Verify that a new user can register successfully
Input	Valid name, email, password
Expected Result	User account is created successfully
Actual Result	User account created successfully
Status	Pass

2. User Login Test Case

Test Case ID	TC_02
Test Case Name	User Login
Description	Verify user can login with valid credentials
Input	Valid email and password
Expected Result	User is redirected to dashboard
Actual Result	User redirected to dashboard
Status	Pass

3. Invalid Login Test Case

Test Case ID	TC_03
Test Case Name	Invalid Login
Description	Verify system rejects invalid credentials
Input	Incorrect email or password
Expected Result	Error message is displayed
Actual Result	Error message displayed correctly
Status	Pass

4. Join Class Test Case

Test Case ID	TC_04
Test Case Name	Join Class
Description	Verify student can join a scheduled class
Input	Click "Join Class"
Expected Result	User enters live session
Actual Result	User successfully entered live session
Status	Pass

5. Create Class Test Case (Instructor)

Test Case ID	TC_05
Test Case Name	Create Class
Description	Verify instructor can create a class
Input	Class name, date, time
Expected Result	Class is created and displayed
Actual Result	Class created and displayed successfully
Status	Pass

6. Schedule Session Test Case

Test Case ID	TC_06
Test Case Name	Schedule Session
Description	Verify instructor can schedule a session
Input	Date and time
Expected Result	Session is scheduled successfully
Actual Result	Session scheduled successfully
Status	Pass

7. Conduct Live Class Test Case

Test Case ID	TC_07
Test Case Name	Conduct Live Class
Description	Verify instructor can start live session
Input	Start session button
Expected Result	Live video session starts
Actual Result	Live session started successfully
Status	Pass

8. Logout Test Case

Test Case ID	TC_08
Test Case Name	Logout
Description	Verify user can logout successfully
Input	Click logout
Expected Result	User is logged out and redirected
Actual Result	User logged out and redirected successfully
Status	Pass

6.4 VALIDATION

Validation is a crucial phase in the development of the Live Interactive Dance Learning Platform, ensuring that the system meets the intended requirements and performs effectively in real-world scenarios. The validation process focuses on verifying whether the developed system satisfies user needs, provides accurate outputs, and maintains reliability during execution. It involves evaluating the overall functionality, usability, performance, and consistency of the application.

In this project, validation was carried out by testing the system under different conditions and with multiple user roles, including students and instructors. Each module of the system, such as user registration, login authentication, class scheduling, and live session connectivity, was validated individually and then collectively to ensure seamless integration. The system was

checked for correctness of data processing, ensuring that user inputs are properly handled and stored in the database without any loss or inconsistency.

Special attention was given to the real-time communication feature implemented using WebRTC technology. The live video session module was validated by conducting multiple trial sessions to ensure stable connectivity, minimal latency, and proper synchronization between participants. The audio and video transmission quality were observed to confirm that the platform delivers a smooth and interactive learning experience similar to physical dance classes.

The user interface was also validated to ensure ease of use and accessibility. The platform was tested across different devices and browsers to confirm responsiveness and compatibility. Navigation between pages, clarity of instructions, and user interaction elements were carefully evaluated to ensure that even first-time users can operate the system without difficulty.

Error handling mechanisms were validated by providing invalid inputs and observing the system's response. The system successfully displayed appropriate error messages and prevented unauthorized access, ensuring security and data integrity. Additionally, database validation was performed to confirm that all records related to users, classes, and participation are accurately maintained.

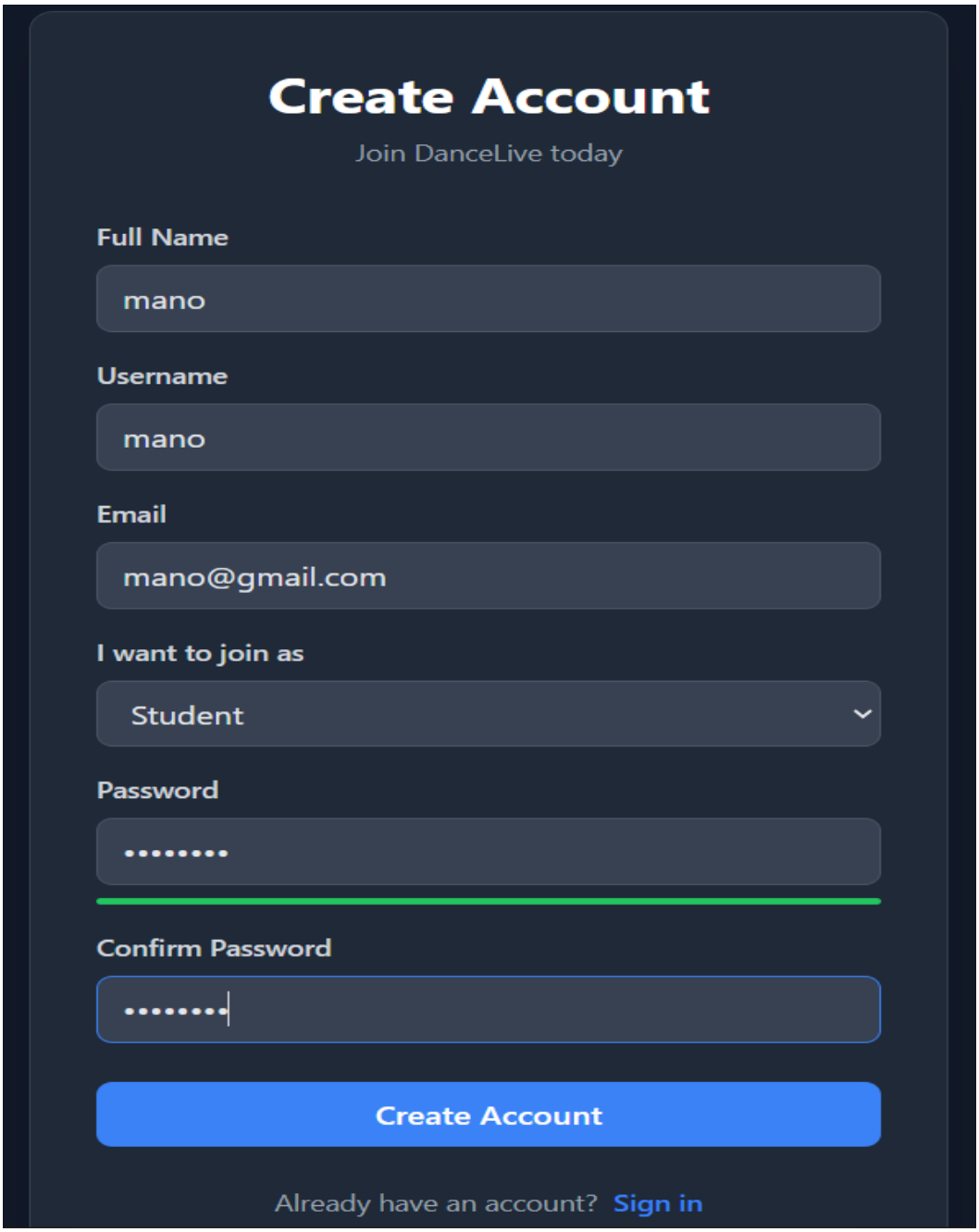
Performance validation was conducted by simulating multiple users accessing the system simultaneously. The platform demonstrated stable performance without significant delays or crashes, indicating its capability to handle real-time usage effectively.

Overall, the validation process confirms that the Live Interactive Dance Learning Platform functions as expected and meets the objectives defined during the system analysis phase. The system is reliable, user-friendly, and capable of delivering an efficient online dance learning experience.

7. USER MANUAL

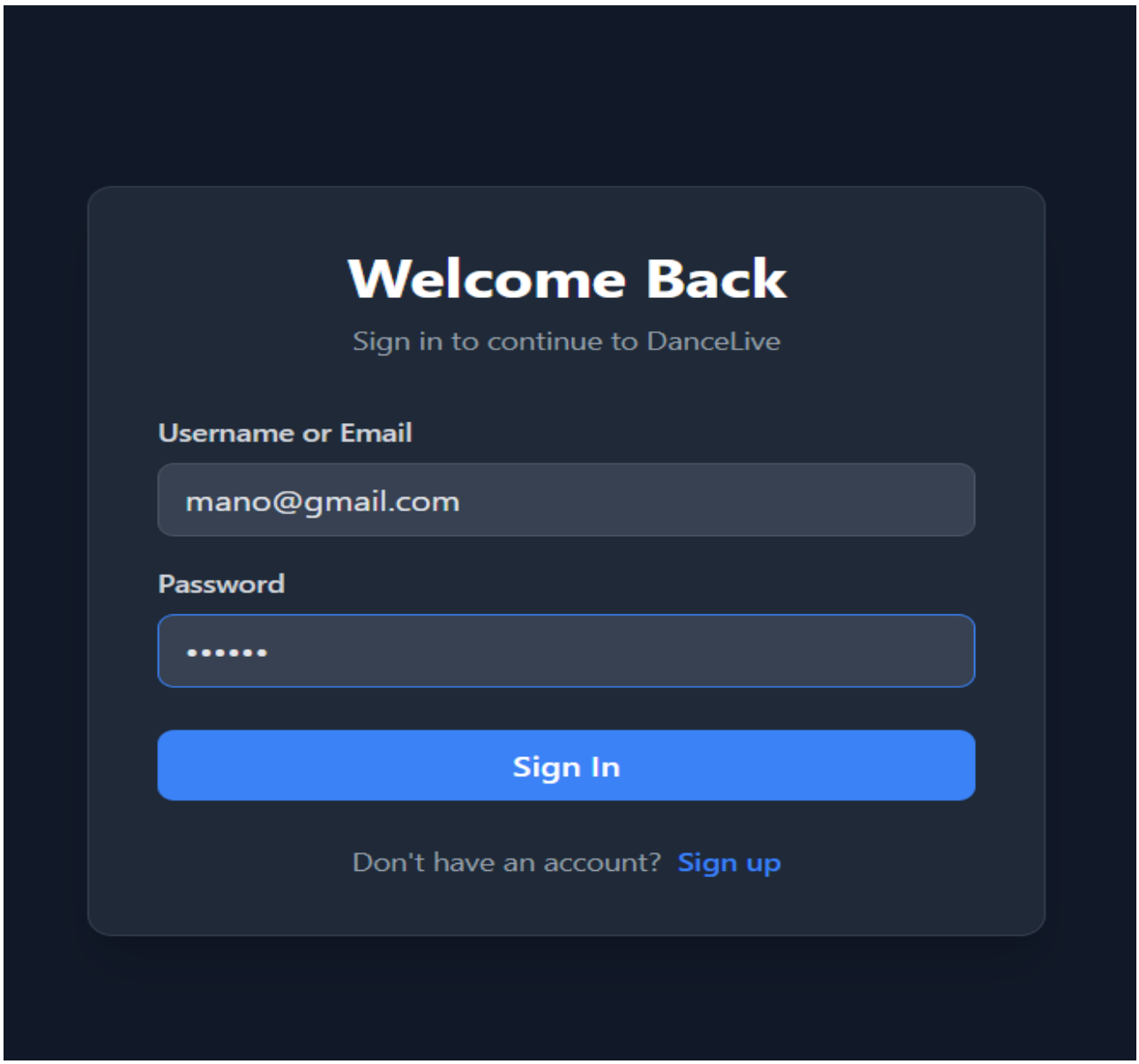
Sign up:

Sign Up is the process through which a new user registers into the Live Interactive Dance Learning Platform by providing required details such as name, email, and password. This process creates a unique account for the user, allowing them to access the system's features. Once the registration is completed successfully, the user can log in using the created credentials.



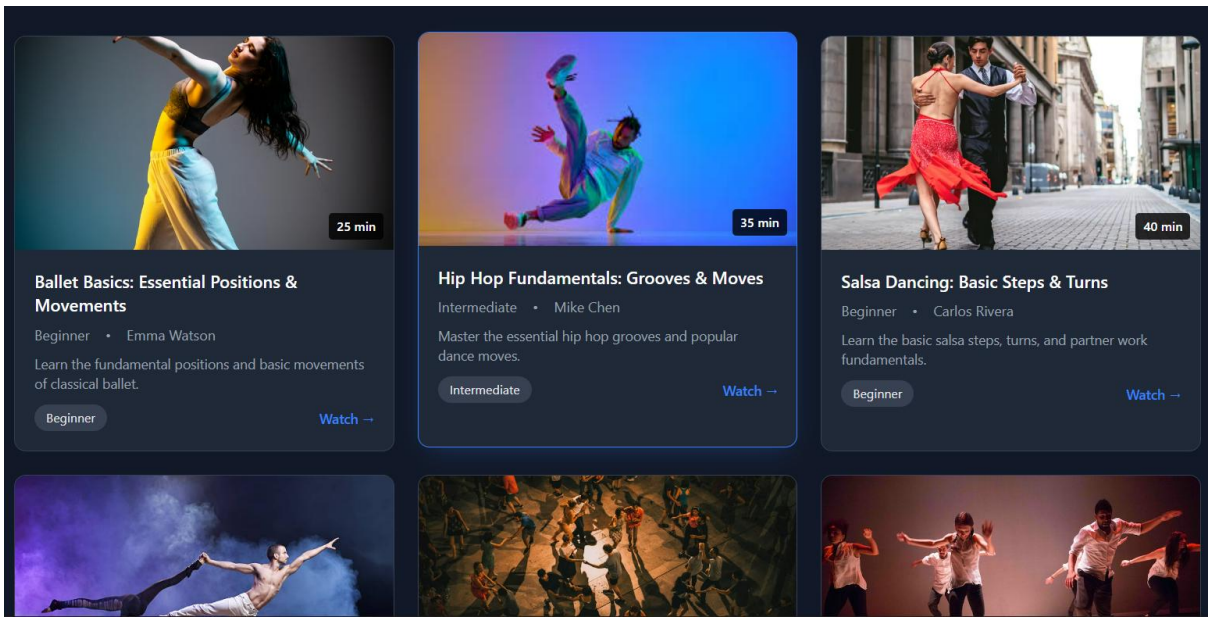
Sign in :

Sign In is the authentication process that allows registered users to access their accounts by entering valid login credentials such as email and password. Upon successful verification, the user is redirected to the dashboard, where they can access various functionalities like joining classes and attending sessions. This ensures secure and authorized access to the system.



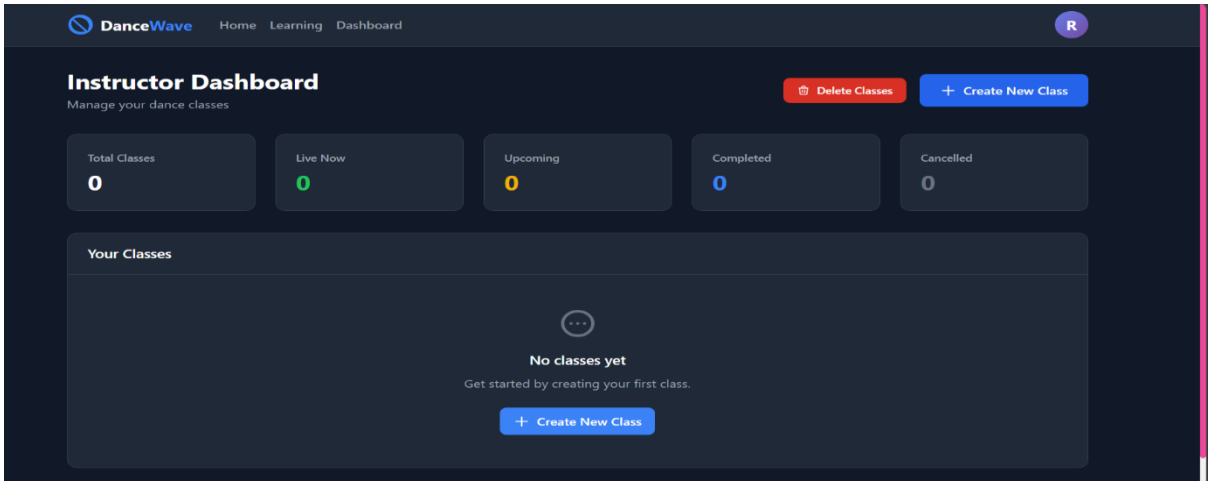
Learning Videos :

The Learning Videos feature allows users to access pre-recorded dance tutorials provided by instructors. These videos help students learn at their own pace by watching and practicing dance steps multiple times. It enhances understanding and acts as a supplementary learning resource alongside live sessions.



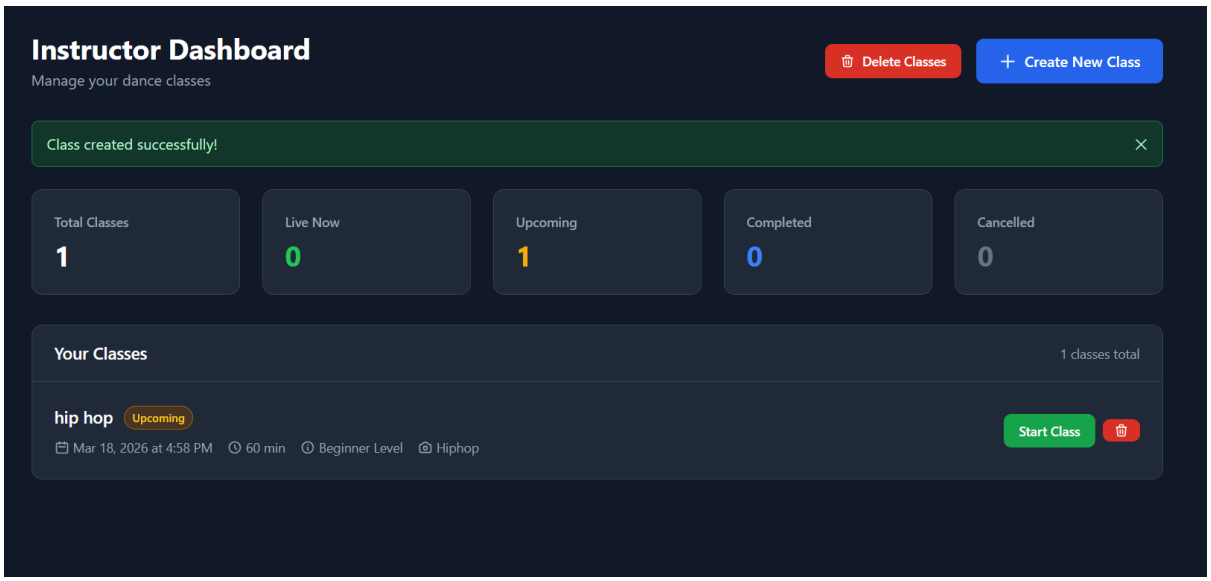
Instructor Dashboard :

The Instructor Dashboard is the main interface provided to instructors after logging into the system. It displays all essential controls and information such as created classes, scheduled sessions, and participant details. This dashboard helps instructors manage their teaching activities efficiently in a structured manner.



Instructor Created Class :

The Instructor Created Class feature allows instructors to create new dance classes by entering details such as class name, type of dance, and session description. Once created, these classes are made available for students to view and join. This feature helps instructors organize and manage different sessions effectively.



Video Call of a meeting :

The Video Call of Meeting feature enables real-time interaction between instructors and students through live video sessions. It allows participants to see and communicate with each other, creating a virtual classroom environment similar to physical dance training. This feature enhances engagement and provides immediate feedback during learning.

8. SYSTEM DEPLOYMENT

Implementation refers to the process of executing a planned system so that it functions effectively in its intended environment. In software development, implementation includes coding, integrating different modules, configuring system components, testing functionalities, and running the application. It ensures that the developed system performs according to the specified requirements and delivers the expected output.

System deployment is the process of making the developed application available for real-time usage. It involves setting up the server environment, configuring databases, ensuring connectivity, and testing the application under real-world conditions. The main objective of deployment is to ensure that the system operates smoothly for end users with reliability and minimal errors.

For the **Live Interactive Dance Learning Platform**, deployment ensures that instructors and students can access the platform seamlessly to create, schedule, and participate in live dance classes. The system enables real-time interaction through video communication, allowing participants to view each other's video streams similar to video conferencing platforms.

Technology Stack Used for Implementation

The Live Interactive Dance Learning Platform is developed using HTML, CSS, PHP, JavaScript, MySQL, and WebRTC. HTML and CSS are used to design a responsive and user-friendly interface, while JavaScript enables interactivity and real-time communication features. PHP handles server-side processing, including user authentication and class management. MySQL is used as the backend database to store user details, class schedules, and participation records securely. WebRTC is integrated to enable live video streaming between instructors and students. This technology stack ensures efficient performance, scalability, and real-time interaction, providing a seamless and engaging online dance learning experience.

Deployment Process and Techniques

The deployment process includes designing the GUI, backend development, database configuration, and server setup. The interface is created using HTML, CSS, and JavaScript to ensure responsiveness across devices. The backend is developed using PHP, which manages user roles, class scheduling, and session handling. MySQL is configured to store and manage system data efficiently. WebRTC is implemented for real-time video communication. The

system is deployed using XAMPP or a web server, ensuring proper connectivity and performance. Testing and debugging are conducted to ensure system reliability. Post-deployment activities include monitoring, maintenance, and updates to improve system functionality and user experience.

3. Server-Side Implementation

• The backend system is implemented using PHP, which manages user authentication, role-based access control, and class creation.
• Instructors are given permission to create and schedule classes, while students can join sessions using meeting links.
• Session management is used to maintain user login states securely.
• Security measures such as password hashing, input validation, and role-based authorization are implemented.

4. Real-Time Communication Setup (WebRTC Integration)

• WebRTC is used to establish peer-to-peer video communication between participants.
• A signaling mechanism (using JavaScript and WebSocket/Socket.IO if integrated) is implemented to exchange connection data such as SDP and ICE candidates.
• STUN servers are used for NAT traversal, and TURN servers can be configured for reliable connectivity in restricted networks.
• Each participant's video stream is displayed on the screen, enabling real-time interaction similar to Google Meet.

5. Testing and Debugging

• The system undergoes multiple testing phases including unit testing, integration testing, and system testing.
• Real-time communication is tested across different network conditions to ensure stable video streaming.
• Debugging tools and browser developer tools are used to identify and fix errors.
• Performance testing ensures the system can handle multiple users in a live session.

6. Packaging and Compression

• The complete project, including PHP files, frontend assets, and database scripts, is packaged into a compressed file for easy distribution.
• Configuration files are included for quick setup of the application environment.
• Database backup files are provided for easy restoration on deployment servers.

7. Deployment on a Web Server

• The application is deployed on a local server environment such as XAMPP or on a live hosting server supporting PHP.
• The MySQL database is configured and connected to the application.
• A domain or local server address is used to allow users to access the platform.
• HTTPS is recommended for secure communication, especially for WebRTC functionality

8. System Maintenance and Updates

• The system is monitored regularly after deployment to ensure stable performance.
• Bug fixes and updates are applied periodically to improve functionality and security.
• Feedback from users (students and instructors) is collected to enhance usability.

HARDWARE RESOURCE

• Processor : Dual Core Processor 2.6 GHz
• RAM : 4 GB
• Hard Disk : 320 GB
• Internet Connection : Required for live video communication
• Operating System : Windows 10 or above

9. CONCLUSION

The Live Interactive Dance Learning Platform has been successfully designed and developed to provide an efficient and modern solution for online dance education. This project effectively addresses the limitations of traditional dance learning methods, such as geographical restrictions, lack of flexibility, and limited access to professional instructors. By integrating real-time communication technologies, the system creates an interactive virtual learning environment that closely resembles physical dance classes.

The platform enables instructors to conduct live sessions and allows students to participate actively through video interaction. This real-time engagement improves learning outcomes by allowing immediate feedback, correction of mistakes, and better understanding of dance techniques. The use of modern web technologies such as HTML, CSS, JavaScript, PHP, MySQL, and WebRTC ensures that the system is reliable, scalable, and capable of handling multiple users simultaneously.

The system also provides essential features such as user authentication, role-based access, class scheduling, and participation management. These features contribute to an organized and structured learning process for both instructors and students. The responsive user interface ensures accessibility across various devices, including desktops, tablets, and smartphones, making the platform convenient and user-friendly.

During the development of this project, various stages such as system analysis, system design, implementation, and testing were carefully carried out. Testing methods including unit testing, integration testing, and system testing were used to ensure the system functions correctly and meets user requirements. The deployment process ensured that the platform operates efficiently in a real-time environment with proper performance and security measures.

Overall, the Live Interactive Dance Learning Platform demonstrates the effective use of web technologies to enhance online education. It provides a flexible, interactive, and accessible learning solution that benefits both students and instructors. This project not only improves the quality of dance education but also showcases the potential of real-time communication systems in transforming traditional learning methods into digital experiences.

10. FUTURE ENHANCEMENT

The Dance Learning Platform can be further enhanced with several advanced features to improve functionality and scalability.

One major enhancement would be the integration of class recording, allowing students to revisit previous sessions for practice and revision.

Another improvement could include screen sharing and multi-camera support, enabling instructors to demonstrate complex dance movements more effectively.

The system can also be upgraded with payment integration, allowing students to enroll in paid courses and instructors to monetize their classes.

Advanced features such as attendance tracking and performance analytics can help instructors monitor student progress and provide personalized feedback.

Integration of chat and messaging systems can improve communication between instructors and students during and after sessions.

Additionally, the system can be extended to support large-scale classes using SFU architecture, enabling more participants without performance issues.

A mobile application for Android and iOS can also be developed to improve accessibility and user convenience.

These enhancements will make the system more powerful, scalable, and suitable for real-world deployment.

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APPENDIX A – DATA DICTIONARY

Users :

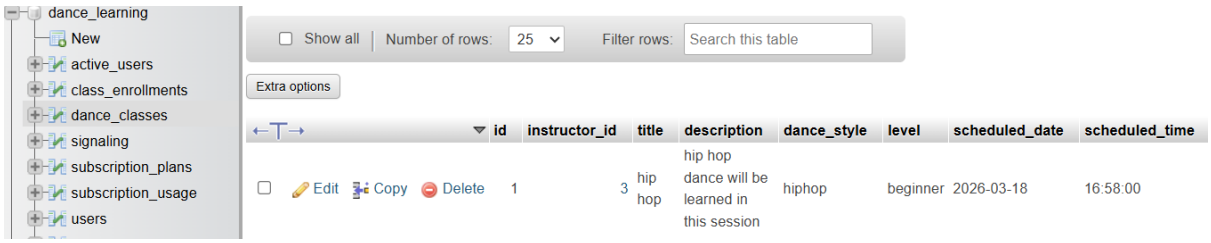
The User table stores the information of all registered users in the system. It includes both students and instructors who access the platform. This table contains details such as user ID, name, email, password, and role. It is mainly used for user authentication and to control access to different features of the system based on the user's role.

		id	username	email	password	role
Edit Copy Delete		1	raghul	raghul@gmail.com	$2y$10$jO1HsF7144.r/KNSq9TRVu.XlfybDuGBlWFIZwNrAth...	student
Edit Copy Delete		2	yazhini	yazhini@gmail.com	$2y$10$Ti7Tp4Hz7xCKnrMR2nzNYu2XUNj/CQMjVHBRvUf1mr....	student
Edit Copy Delete		3	rag	rag@gmail.com	$2y$10$UnjLll1lez6cNycnimufbuEVAeXShJ2HG11KgYwb2e7...	instructor

Dance Meetings :

The Dance Meeting table maintains information about all live dance sessions conducted on the platform. It includes details such as meeting ID, instructor ID, scheduled date and time, and meeting link. This table is used to organize and manage live classes, allowing students to join sessions and instructors to conduct them efficiently.

	id	instructor_id	title	description	dance_style	level	scheduled_date	scheduled_time
Edit Copy Delete	1	3	hip hop	hip hop dance will be learned in this session	hiphop	beginner	2026-03-18	16:58:00



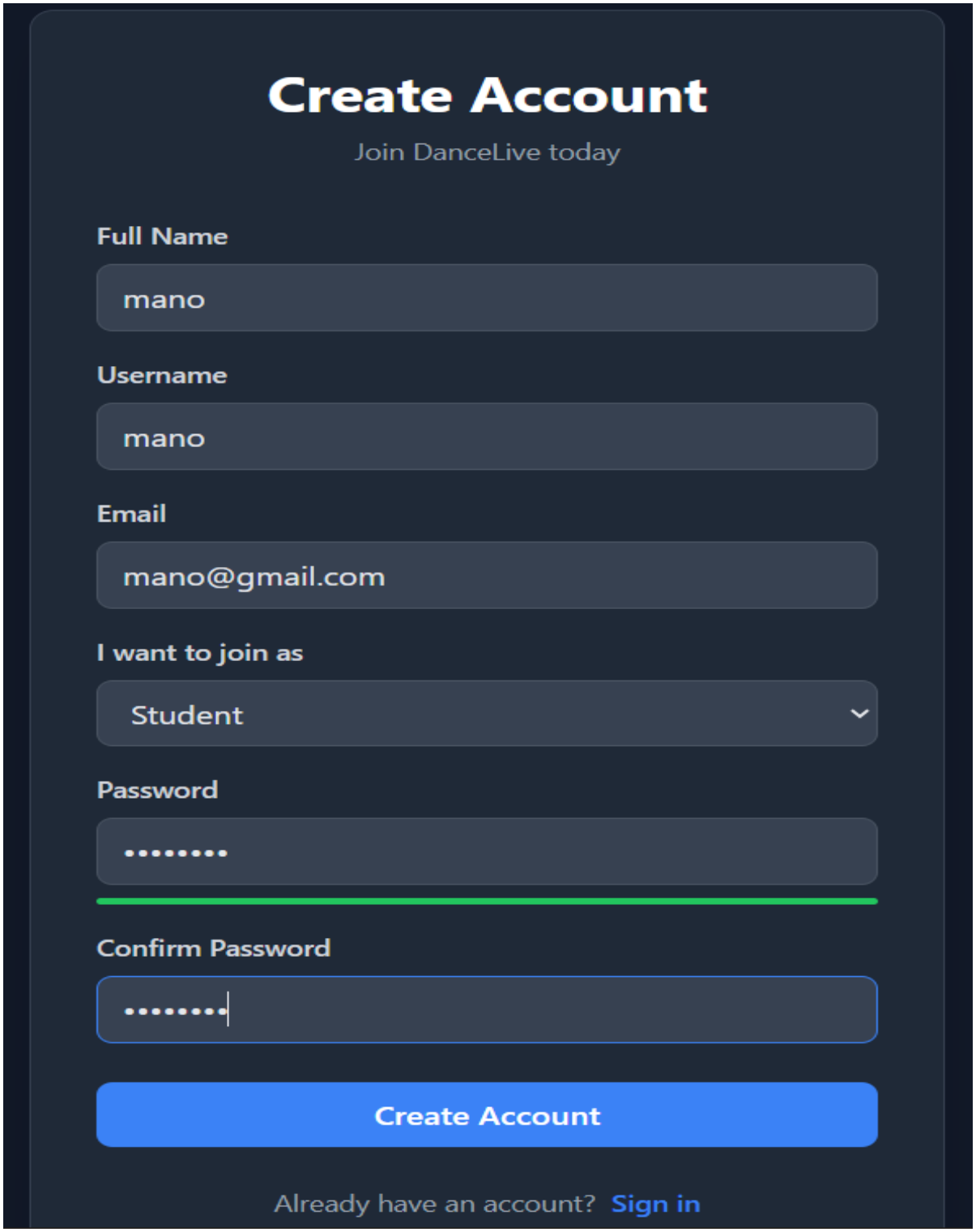
Signals :

The Signals table is used to manage real-time communication between users during live sessions. It stores signaling data required for establishing connections using WebRTC technology. This includes session identifiers, sender and receiver information, and message data such as offer, answer, and ICE candidates. This table ensures smooth and stable communication for live video streaming between participants.

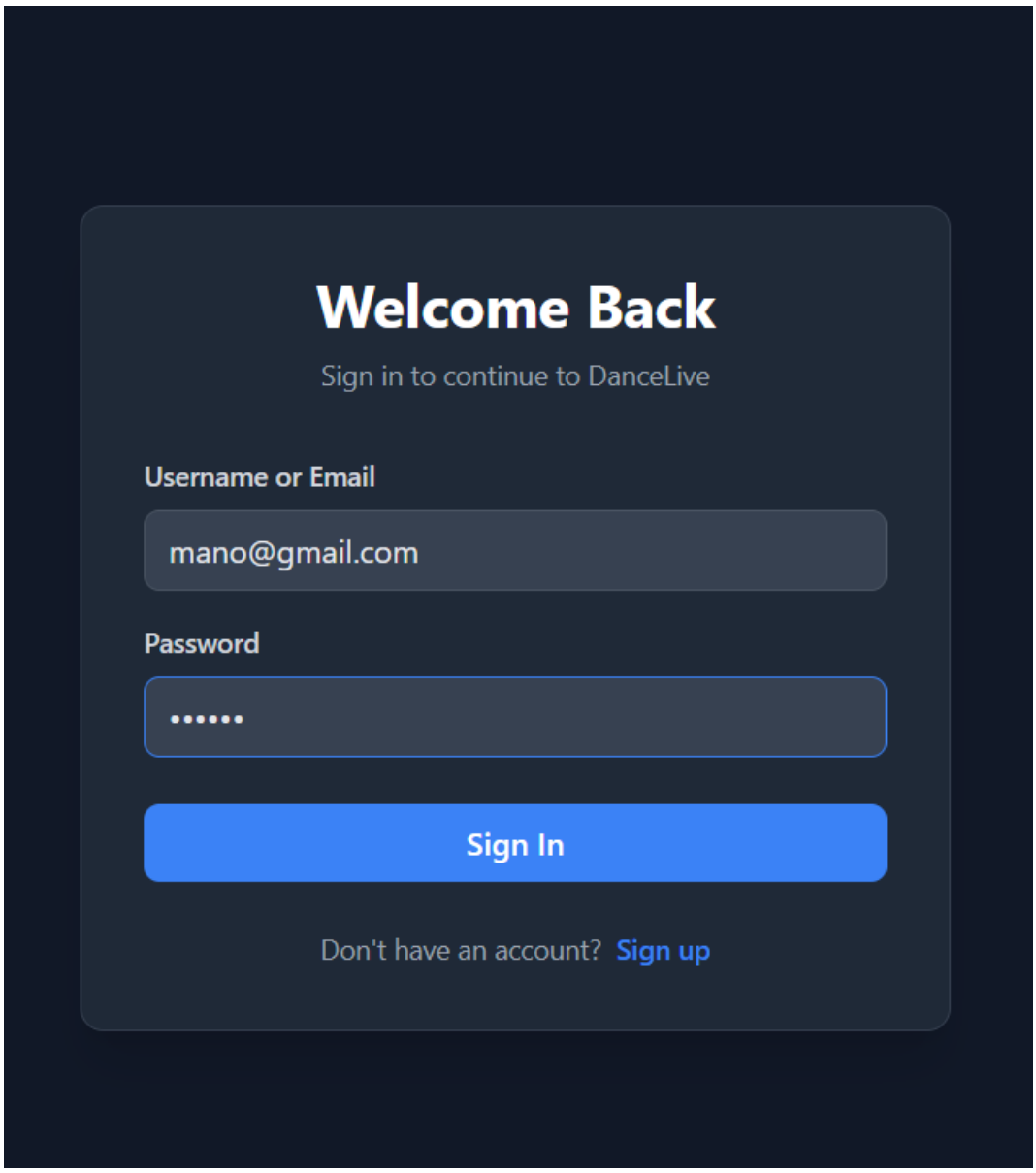
	id	room_id	user_id	user_role	message_type	message	target_user_id	crea
Edit Copy Delete	2	1	3	instructor	leave	{"userId":3}	NULL	2026

APPENDIX B- SAMPLE OUTPUT

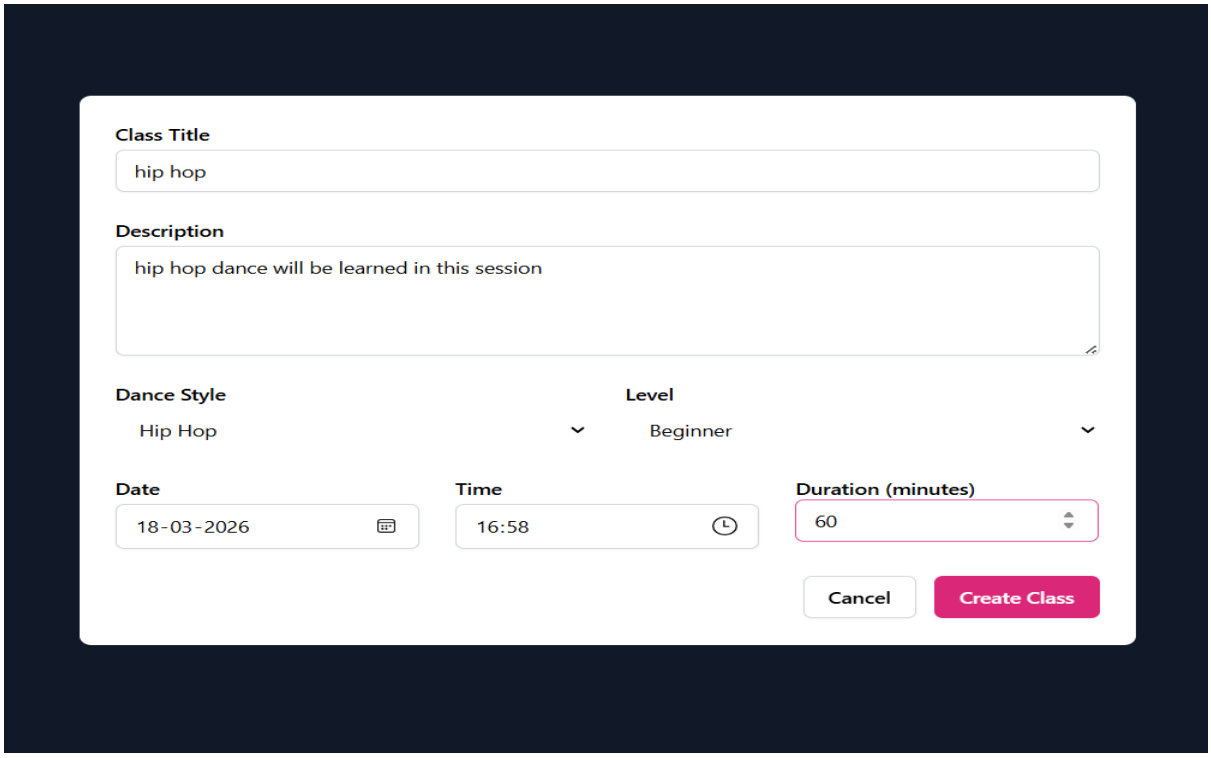
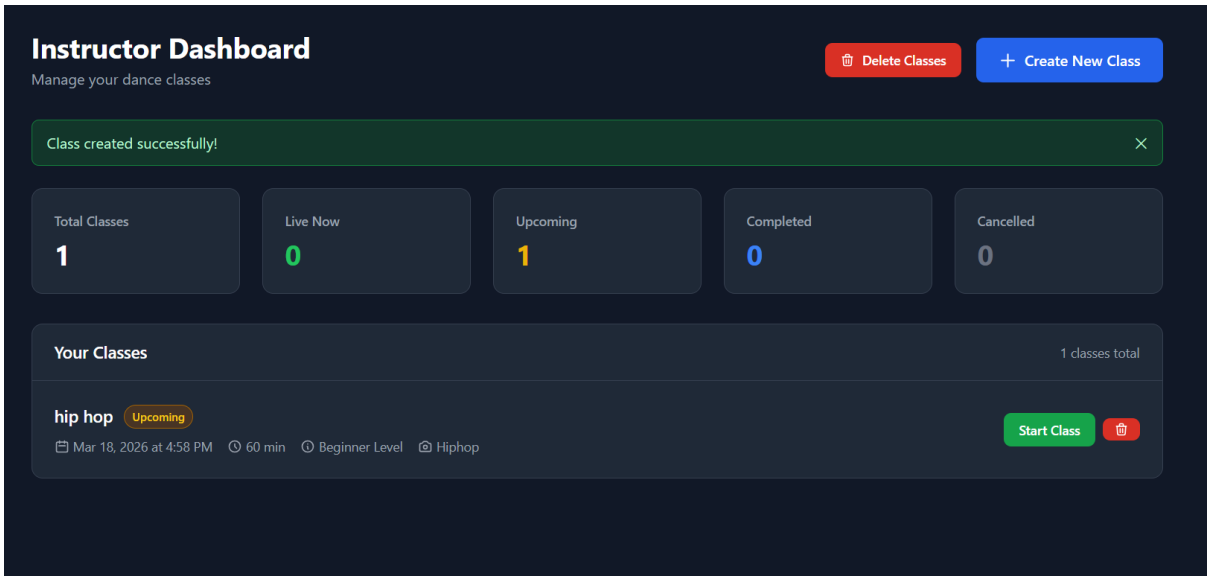
Signup :



Sign in :


Welcome Back
Sign in to continue to DanceLive
Username or Email
mano@gmail.com
Password
••••••
Sign In
Don't have an account? Sign up

Creating Dance Meeting :



Instructor Dashboard :